

Empirical Proof Record: VALIDATED

2026-05-14T20:39:41.476793+00:00

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Proof ID: 9989f42bc8f2b0d230ddc9b61a2d6ff7...

Created: 2026-05-14T20:39:41.476793+00:00

1. Claim

A 50,000-cycle component-level catastrophic run against Kimera-SWM completes within 4.0 hours on this vessel.

- **Operationalisation:** measured steady-state cycles/sec (construction excluded) of the fastest component target, projected to 50000 cycles
- **Threshold:** `projected_50k_component_run_hours` $\leq$ 4.0 hours
- **H0:** projected run time $>$ 4.0h — in-session density testing impractical
- **H1:** projected run time $\leq$ 4.0h — in-session density testing is practical

2. Verdict

VALIDATED — observed 0.5681984997257435
observed 0.568198 satisfies the pre-registered threshold (`projected_50k_component_run_hours` $\leq$ 4.0 hours)

3. Evidence

Pillar	Statistic	Value	95% CI	Library
<code>throughput.entity_cpu</code>	<code>steady_state_cycles_per_sec</code>	4.2772529758101285	—	ophamin
<code>throughput.entity_gpu</code>	<code>steady_state_cycles_per_sec</code>	4.306903481687651	—	ophamin
<code>throughput.arachne_cpu</code>	<code>steady_state_cycles_per_sec</code>	24.205516857317008	—	ophamin
<code>throughput.arachne_gpu</code>	<code>steady_state_cycles_per_sec</code>	24.443726788424712	—	ophamin
<code>throughput.rosetta_cpu</code>	<code>steady_state_cycles_per_sec</code>	24.0279083175282	—	ophamin

4. Pre-registration

- Registered at: 2026-05-14T20:38:58.421081+00:00
- Config hash: 8d26f6573f75b50b4f74150a9a35aec6...
- Data hash: 83109a27c3df2a45b912398b969fa8f7...

5. Signature

3f745dee3a523f4e83c4277c527c51c791d130de2961ef7d713d7c6c90937f5e